

THE UNIVERSITY *of* EDINBURGH

Statistical Case Studies

Project 1, part 1: Uncertainty in river flow hydrographs for flood inundation modelling

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1 Executive Summary

Current flood models assume flood patterns are fixed, adjusting only for maximum flow. This report investigates the uncertainty in both the peak flow rate and the time it takes to reach that maximum from the start of the rise.

Using 10 years of river flow data from the River Don at Parkhill gauging station, we identified 41 major flood events. Both factors varied significantly, with median values of c.104 m^3/s and c.73 hours respectively. Floods primarily occurred in Autumn and Winter, with Autumn events typically taking longer to reach their peak.

We found that larger floods generally take longer to reach maximum flow, though this timing varies considerably between events. Assuming all floods follow a standard pattern may overlook real-world uncertainty. Consequently, SEPA should factor in both peak flow rate and time to peak when assessing flood risk.

2 Data and Exploratory Analysis

2.1 Data Description and Quality

Table 1: Data used in this analysis, and the quality codes assigned to each reading

Data description	
Station	Don at Parkhill
Source	SEPA time series data service
Measurement	River discharge (m^3/s)
Catchment area	1,273 km^2
Resolution	15 minutes
Station type	Velocity-area
Period	01 October 2016 – 30 September 2026
Readings	350,592
Missing readings	0
Quality codes	
50 – Good (G)	335,647 readings (95.74%), retained
100 – Estimated (E)	14,915 readings (4.25%), retained
140 – Provisional (Prov)	0 readings (0.00%), excluded
150 – Suspect (S)	0 readings (0.00%), excluded
200 – Unchecked (imported from legacy database) (V)	0 readings (0.00%), excluded
254 – Unchecked (U)	30 readings (0.01%), excluded

The subsequent analysis uses discharge readings (m^3/s) covering 10 water years. Parkhill, the lowest gauging station on the Don, applies a single peak flow rating curve to all measurements.

Each reading is assigned a SEPA quality code (Table 1).

- Estimated ($n = 14,915$): Retained. SEPA defines these as acceptable rating curve extrapolations where the river level exceeds the 3.7 m gauging limit at which Good readings are capped (Figure 2). These are the largest flow readings, so we retain to preserve the extremes.
- Unchecked ($n = 30$): Excluded. These are restricted to recent entries (29 September 2026, 16:15 onward) and sporadic 2017–2018 occurrences of at most 45 minutes. Only two coincide with extreme flows, so their removal leaves a fully verified data-set with negligible impact.

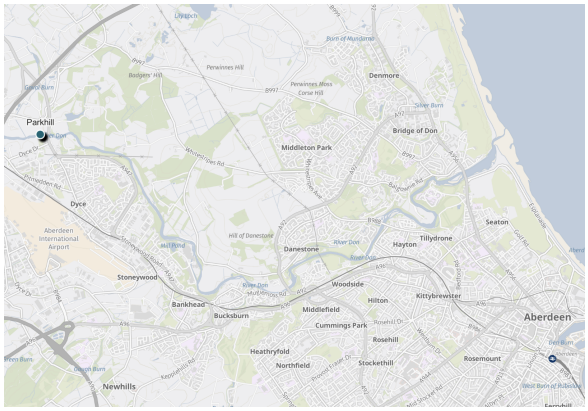


Figure 1: SEPA map of the location of the Parkhill gauging station on the River Don.



Figure 2: Discharge distribution for quality codes 50 and 100.

Figure 3: Discharge distribution for quality codes 50 and 100.

2.2 Report Assumptions

- A *clean peak* is the highest candidate discharge within a 14-day symmetrical window, reliably isolating major flood peaks.
- Start of the rise (t_{rise}):

$$t_{\text{rise}} = \max\{t \in [t_{\text{peak}} - 7 \text{ days}, t_{\text{peak}}) \mid q(t) \leq 1.05 \cdot q_{\text{min}}\}$$

where $q_{\text{min}} = \min_{t \in [t_{\text{peak}} - 7 \text{ days}, t_{\text{peak}})} q(t)$ is the lowest discharge recorded in the week before the peak. The 7-day look-back period aligns well with the synoptic cycles of typical flood-generating storms (3-7 days).

2.3 Exploratory Analysis

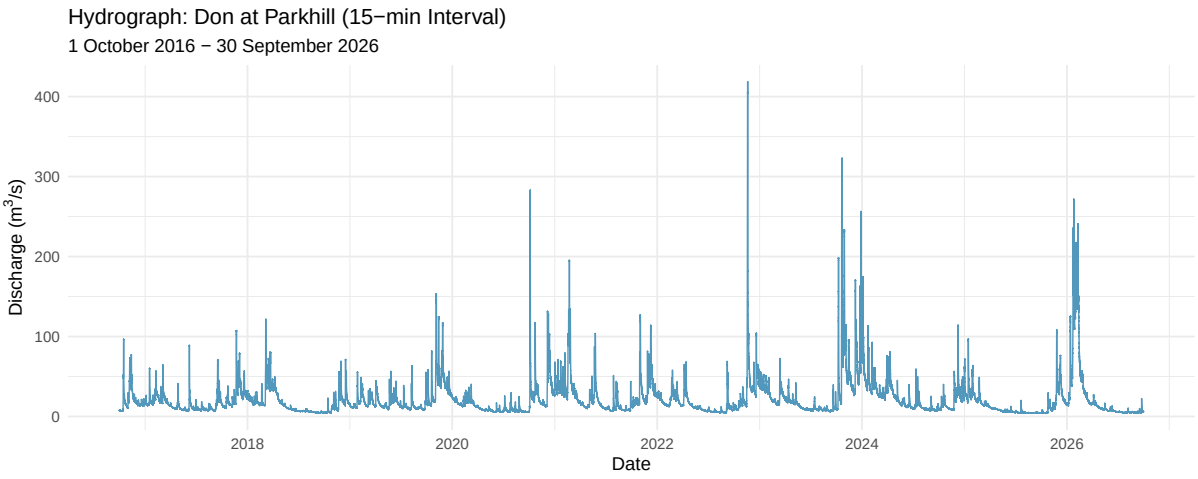


Figure 4: Hydrograph of 15-minute discharge for the Don at Parkhill.

2.3.1 Seasonality of High Flows

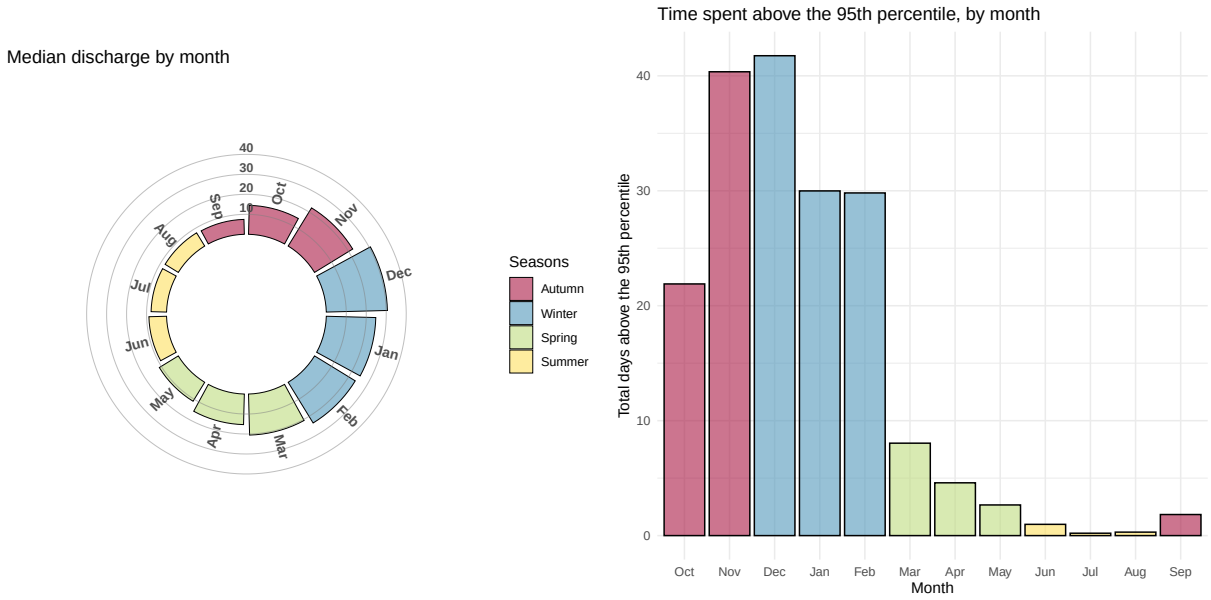


Figure 5: Median discharge by month (left) and total days spent above the 95th percentile in each month (right).

Median discharge is highest from December to February and lowest from July to September. Time spent above the 95th percentile is concentrated between October and February (Figure 5).

2.3.2 Flood Event Hydrographs

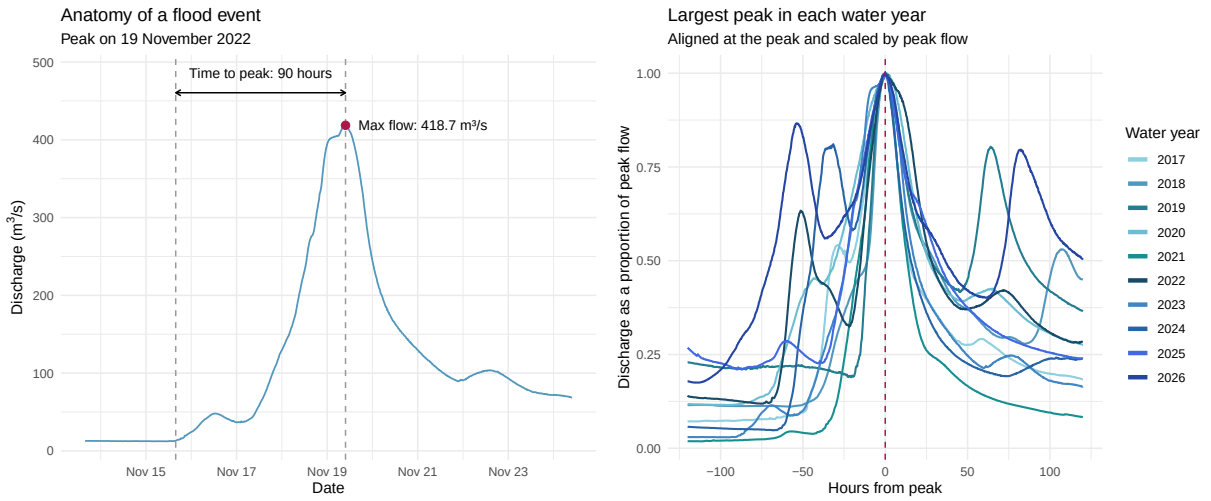
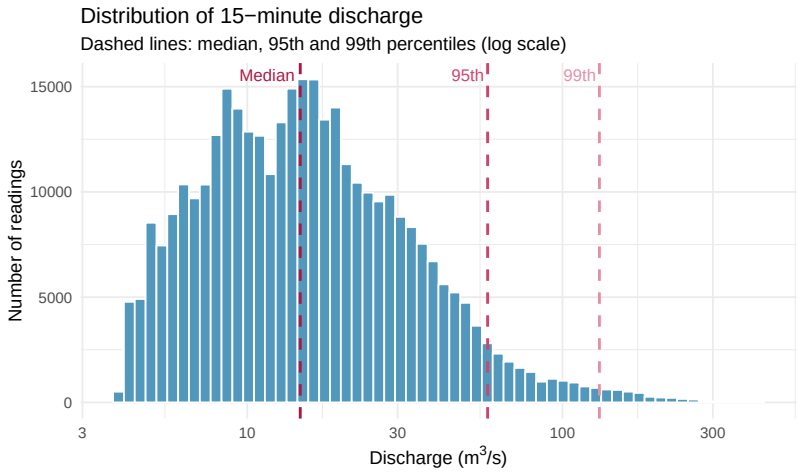


Figure 6: The largest recorded flood event, showing the maximum flow and time to peak calculated using t_{rise} assumption (left), and the largest peak in each water year, aligned at the peak and scaled by peak flow (right).

Figure 6 shows that hydrograph peak shapes vary considerably, and smaller peaks often cluster around maxima meaning extreme flow events must be filtered for independence.

2.3.3 Flow Distribution



Statistic	Value (m³/s)
Mean	21.65
Standard deviation	24.38
Minimum	3.90
25th percentile	8.74
Median	14.72
75th percentile	25.44
95th percentile	57.88
99th percentile	130.96
Maximum	418.68

Figure 7: Distribution of 15-minute discharge on a log scale, with summary statistics. Dashed lines mark the median, 95th and 99th percentiles.

Discharge is strongly right-skewed (Figure 7), with a maximum over 28 times the median, suggesting extreme flows behave differently from typical flows and require separate analysis.

3 Threshold Selection and Independence

3.1 Initial Threshold

We use the Peaks Over Threshold (POT) method to identify extreme flow events exceeding a threshold u, typically set at the 95th percentile or a level yielding 3 to 5 peaks per year. Figure 8 visualises these thresholds: the 95th percentile is 57.9 m^3/s, while raw threshold crossings yield 5, 4, and 3 peaks per year at 79, 93, and 113 m^3/s. However, this raw count overstates distinct events since a single flood can cross the threshold multiple times. Because the 95th percentile yields 8.3 peaks per year, the two rules disagree. Therefore, we follow the peaks-per-year rule – which directly controls event frequency rather than time above threshold – and set our initial threshold to $u = 79$ m^3/s, giving 5 peaks per year, the upper end of the range (Robson and Reed 1999), since independence filtering will reduce this count.

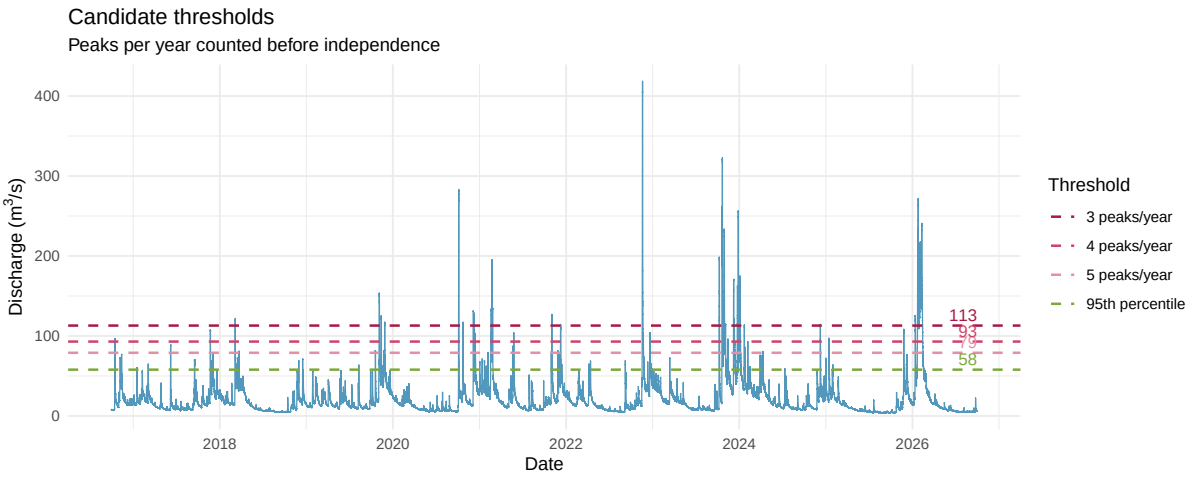


Figure 8: Hydrograph with the candidate thresholds. Peaks per year are counted before applying the independence rules.

3.2 Independence of Peaks

We implement our assumptions to apply the (Robson and Reed 1999) independence rules. If either rule fails, we treat the peaks as one event and keep the larger peak.

This reduces the 47 candidate peaks to 30 independent events, or 3 per year (Figure 9).

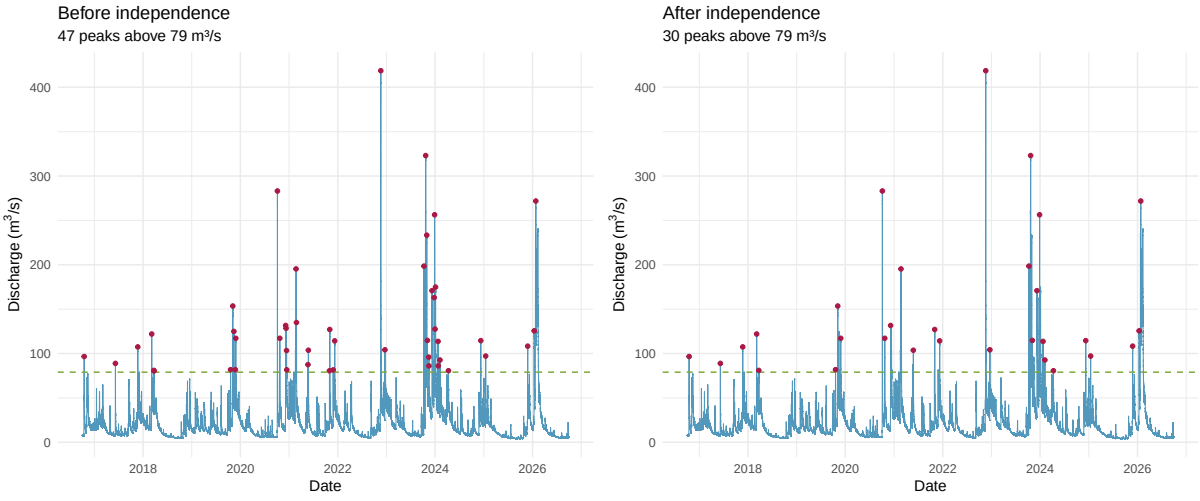


Figure 9: Peaks above the initial threshold before (left) and after (right) applying the independence rules.

3.3 Final Threshold

Our initial threshold yields 3 independent peaks per year, whereas thresholds between 57 and 80 m^3/s achieve the target 3 to 5 peaks per year (Figure 10).

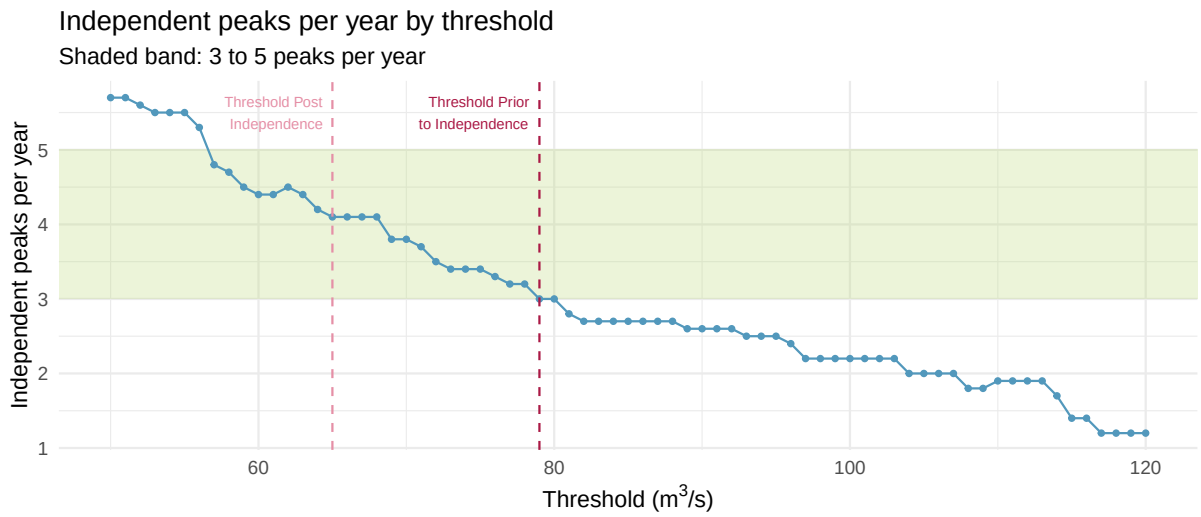


Figure 10: Independent peaks per year at each threshold.

To choose a final threshold, we use the result that excesses over a high enough threshold follow a Generalised Pareto Distribution,

$$P(X - u \leq y \mid X > u) = 1 - \left(1 + \frac{\xi y}{\sigma}\right)^{-1/\xi}, \quad y > 0, \quad 1 + \frac{\xi y}{\sigma} > 0,$$

with scale σ and shape ξ, so that above a valid threshold the mean excess $E[X - u \mid X > u]$ is linear in u and the estimates of ξ and the modified scale, $\sigma - \xi u$, stay roughly constant (Coles 2001). Figure 11 shows that the mean residual life plot is approximately linear and the parameter estimates are roughly constant above 65 m^3/s.

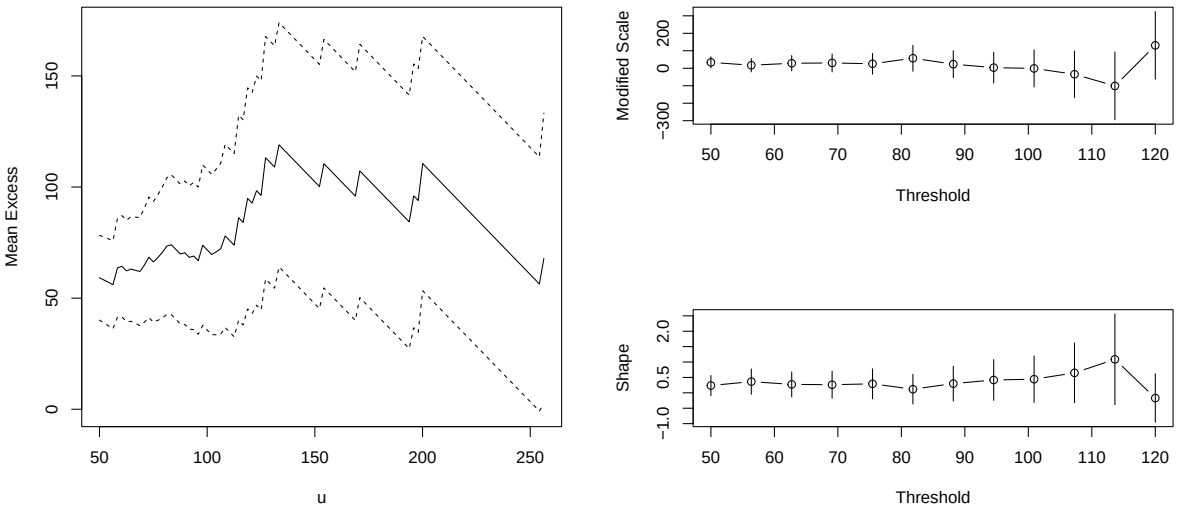


Figure 11: Mean residual life plot (left) and parameter stability plots (right) for independent peaks above 50 m^3/s.

Therefore, we set $u = 65$ m^3/s. Re-applying the independence rules, we obtain 41 independent events (4.1 per year), with fitted parameters $\hat{\sigma} = 49.24$ (standard error 12.6) and $\hat{\xi} = 0.242$ (standard error 0.206).

The diagnostic plots (Figure 12) show a good fit to lower-to-moderate exceedances, but some tail under-estimation and high extrapolation uncertainty at long return periods.

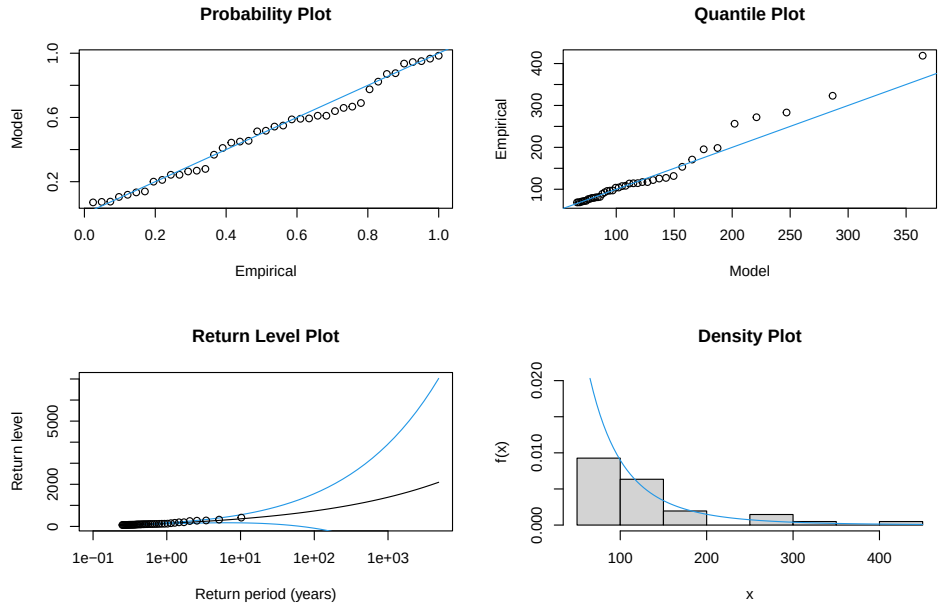


Figure 12: Diagnostic plots for the GPD fitted at the final threshold.

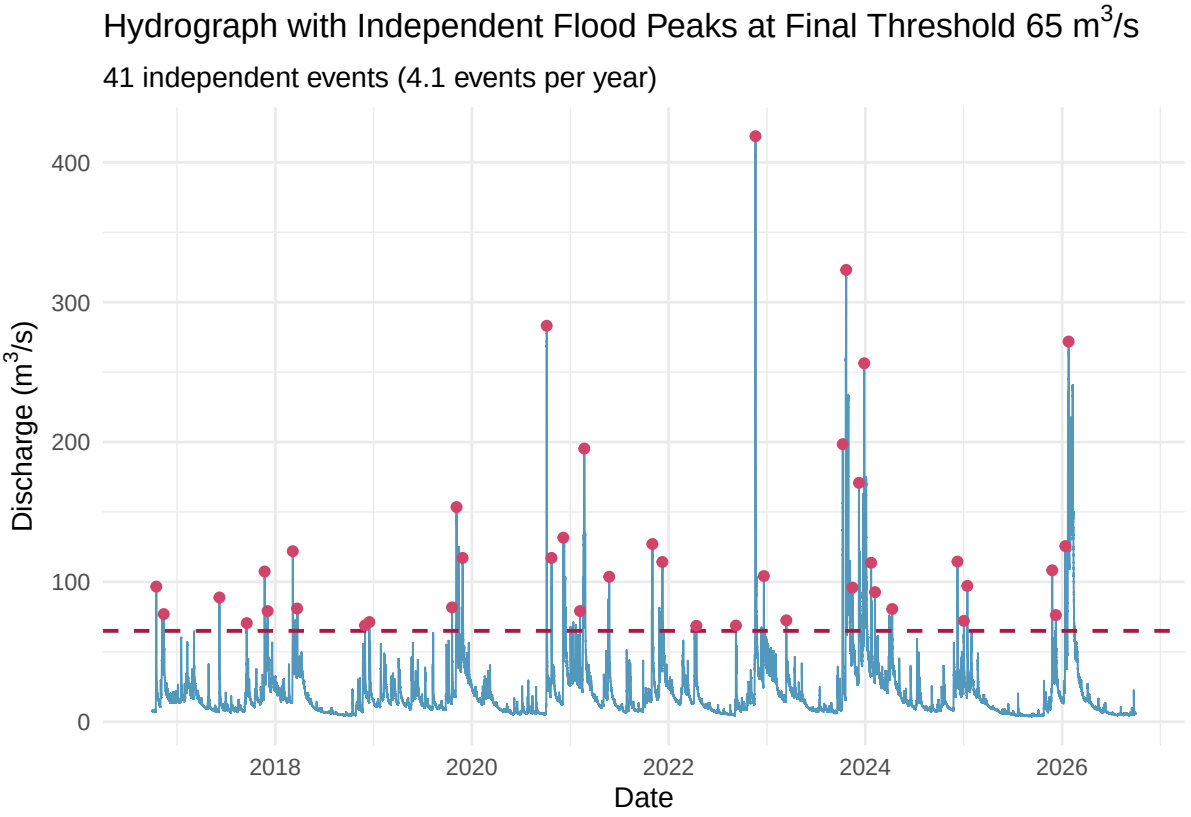


Figure 13: Hydrograph showing the final threshold and each independent flood peak.

4 Uncertainty Analysis

The final threshold of $u = 65$ m^3/s identifies 41 independent flood events. Both maximum flow and time to peak varied considerably (Figures 14 and 15). Maximum flow had a median of 104.21 m^3/s (interquartile range 79.2–127.1 m^3/s), while time to peak had a median of 72.75 hours (interquartile range 38–114 hours) and ranged from 16.8 to 166 hours. This variation suggests that flood events cannot be properly characterised by a single maximum discharge or a fixed time to peak.

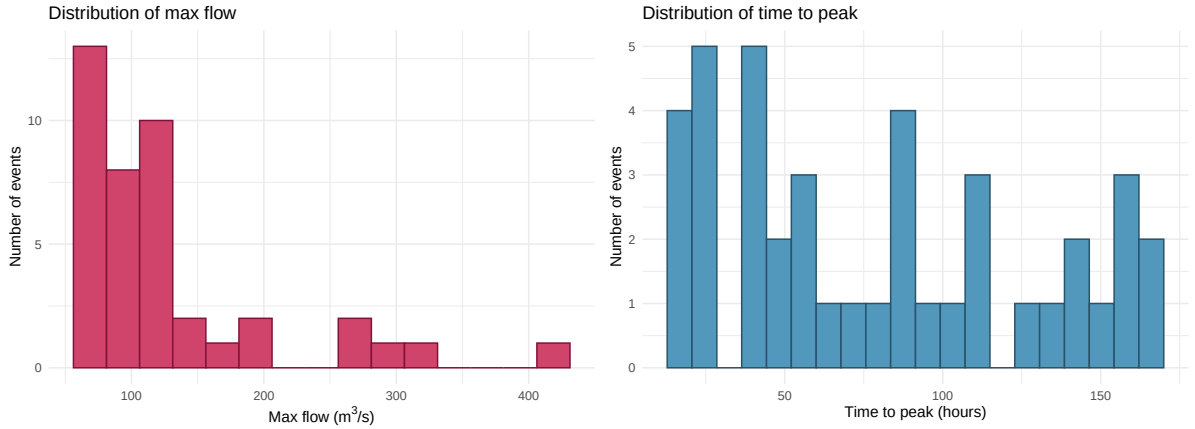


Figure 14: The Distribution of the two parameters

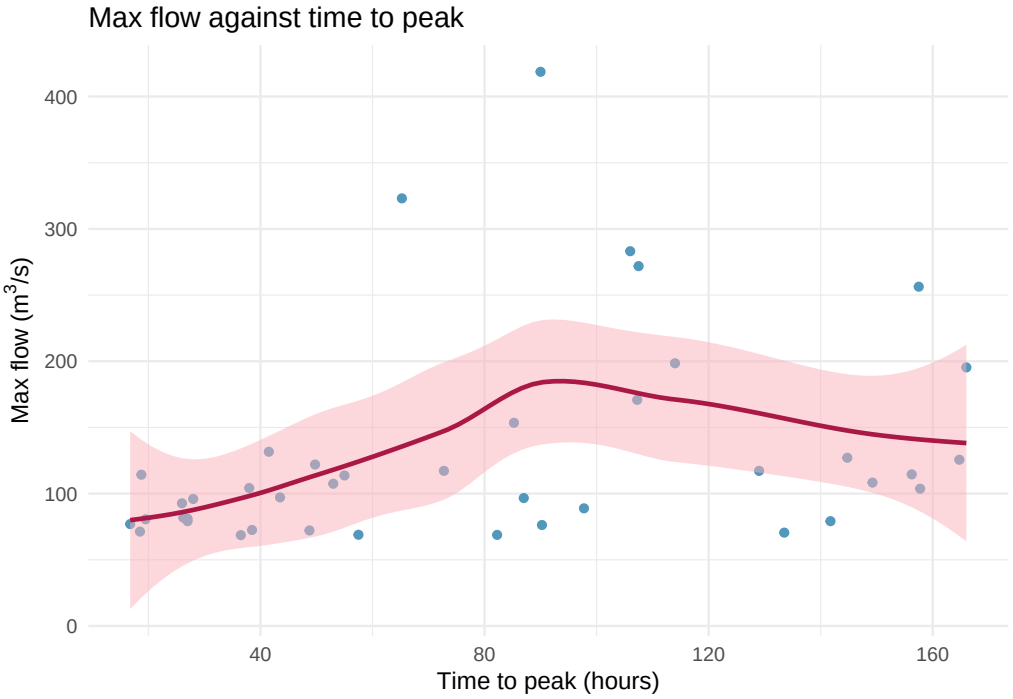


Figure 15: Maximum flow against time to peak, with a smoothed trend

Since both variables are skewed, we use Spearman's rank correlation to investigate a potential monotonic relationship. Table 2 indicates a moderate, statistically significant positive association ($\rho = 0.458$, $p = 0.003$), suggesting that larger flood peaks tend to be associated with longer

times to peak. However, the scatter plot in Figure 15 suggests that peak discharge alone does not determine the time taken to reach the maximum.

Table 2: Spearman's Rank Correlation Between Peak Discharge and Time to Peak

Test	Rho	P-value	S statistic
Spearman's rank correlation	0.458	0.003	6223.813

Table 3: Summary of Max Flow and Time to Peak for each Season

season	Events	Median max flow	Median time to peak (h)
Winter	17	113.694	55.00
Spring	6	80.812	37.50
Summer	1	88.889	97.75
Autumn	17	108.308	85.25

Seasonal differences were also observed (Table 3, Figure 16). Autumn events ($n = 17$) had a median maximum discharge of 108.308 m^3/s and time to peak of 85.25 hours, compared with 113.694 m^3/s and 55.00 hours for Winter events ($n = 17$), highlighting that Autumn floods tend to take longer to reach peak flow. However, the season sample sizes are limited, particularly for Spring ($n = 6$) and Summer ($n = 1$), so these differences should be interpreted cautiously.

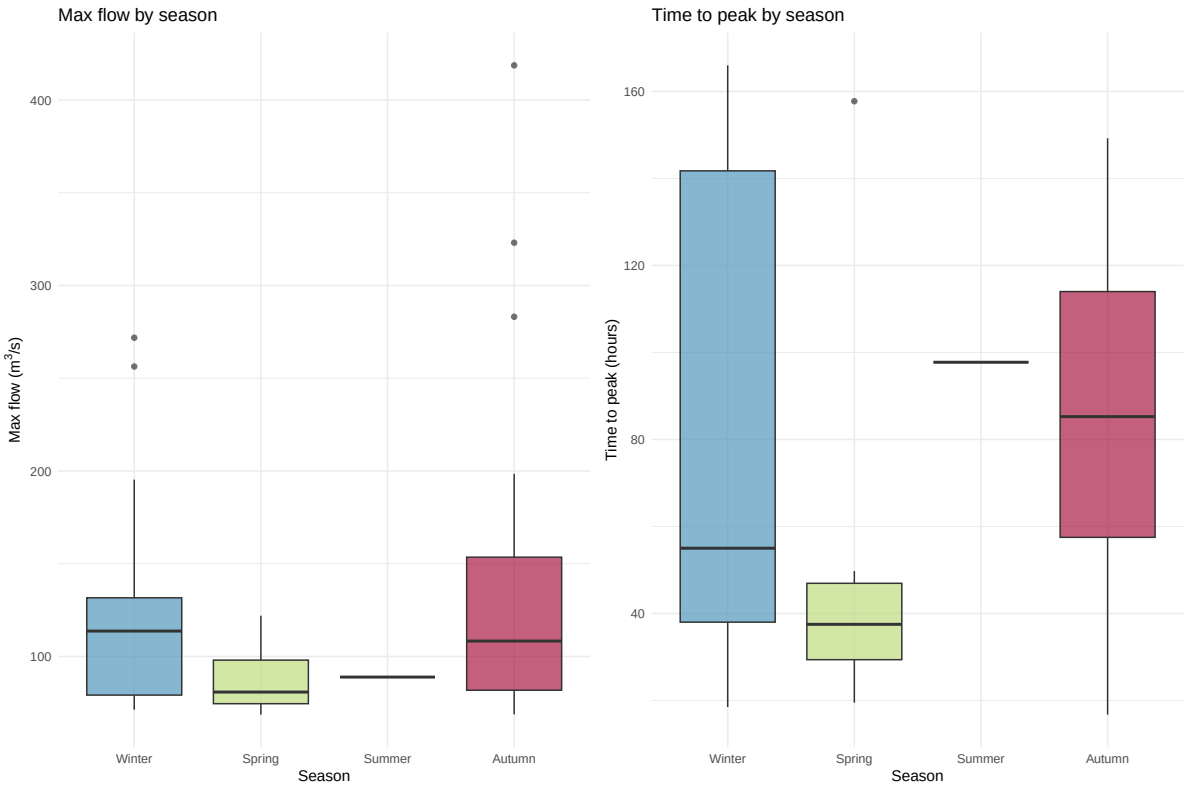


Figure 16: A comparison of the uncertainty in the parameters by season

Repeating the analysis across thresholds of 55-85 m^3/s (Table 4), median time to peak ranges from 65.250 hours to 90.000 hours, while Spearman's correlation coefficient ranges from 0.308

Table 4: Final uncertainty table, with thresholds satisfying 3-5 events per year criteria highlighted.

threshold	events	Events per Year	Median Max Flow	Median Time to Peak (Hours)	Spearman Rho Correlation
55	55	5.5	81.792	65.250	0.308
60	**44**	**4.4**	**100.434**	**77.500**	**0.362**
65	**41**	**4.1**	**104.208**	**72.750**	**0.458**
70	**38**	**3.8**	**107.884**	**79.000**	**0.462**
75	**34**	**3.4**	**113.975**	**86.125**	**0.471**
80	**30**	**3.0**	**115.977**	**86.125**	**0.488**
85	27	2.7	117.185	90.000	0.321

to 0.488. The highlighted thresholds satisfy the 3–5 events-per-year criterion, with the chosen threshold of 65 m^3/s yielding 4.1 independent events per year. Although results vary with threshold selection, a positive correlation between peak discharge and time to peak persists.

References

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